Dataset Options and Updated Dataset Strategy

Fine-Tuning for Guarani Mbya — PRP2

Purpose of this document

This document explains how my dataset strategy developed during PRP2. I started with two options, FLORES+ and AfroLingu-MT, but after investigating FLORES+ more carefully, I found it was not suitable as training data. This led me to Guarani Wikipedia as my final training dataset.

The goal is not only to describe the final choice, but to show the reasoning behind the change.

Project context

My PRP2 project is about fine-tuning a small language model for a low-resource language writing assistant, inspired by Guarani Mbya. I will not use sensitive community data. Instead, I use safe public data to test the core technical question:

> ***Can a small fine-tuned model adapt to limited language data, or does it mostly memorize?***

The dataset needs to be suitable for training, small enough for a low-resource experiment, safe and publicly available, technically usable in Python, splittable into train/validation/test sets, and realistic for a 19-day student project.

Initial dataset options

At the start, I considered two options: FLORES+ and AfroLingu-MT. Guarani Wikipedia was not part of my first options; I found it later after ruling out FLORES+.

Option 1: FLORES+

Link: https://huggingface.co/datasets/openlanguagedata/flores_plus

Type: Parallel multilingual / low-resource translation dataset

Why I first considered it

FLORES+ looked like a good fit because it is already structured, multilingual, and connected to low-resource language work, easy to explain in a portfolio and safe to use without collecting community data. Compared to AfroLingu-MT, it also seemed broader and less complex: AfroLingu-MT required choosing an African language pair and understanding an MT-specific setup, which would add unnecessary scope. At first glance, FLORES+ was the stronger option.

Why I decided not to use it for training

However, after looking more carefully at the dataset documentation, I found that FLORES+ explicitly states it should not be used as training data. The dataset card says: “It should not be used as training data.” This made the decision clear.

Beyond that official statement, the reason also makes sense for my project. FLORES+ was designed to measure translation quality. The data itself consists of English text translated into over 200 language varieties by professional translators, with the original sentences sampled from Wikinews, Wikijunior, and Wikivoyage. This means the Guarani text in FLORES+ is not original Guarani writing. It is translated, formal, and standardized.

My project is not about machine translation benchmarking. It is about testing whether a small model can adapt to real, limited, natural Guarani text. FLORES+ does not serve that goal, not because of its quality, but because it was built for a different purpose entirely. I could technically split it into train, validation, and test sets, but that would not change what kind of text it contains.

Current role of FLORES+

Not used as training data. It remains an optional evaluation reference if it fits the final setup.

Option 2: AfroLingu-MT

Link: https://huggingface.co/datasets/UBC-NLP/AfroLingu-MT

Type: African low-resource machine translation dataset

Why I considered it

AfroLingu-MT fits the low-resource theme and is connected to machine translation research, which made it a reasonable comparison option.

Why I did not choose it

It would require choosing a language pair and shifting the project into an African MT direction, adding extra complexity for a 19-day project and moving away from the Guarani-inspired context. FLORES+ was the simpler first option for those reasons. AfroLingu-MT could be useful in a project specifically focused on African machine translation, but not here.

Dataset comparison

The table below summarizes why FLORES+ and AfroLingu-MT were ruled out and why Guarani Wikipedia became the final choice.

	FLORES+	AfroLingu-MT	Guarani Wikipedia
Languages	200+ including Guarani	African languages only	Paraguayan Guarani only
Data type	Translation pairs (EN → X)	Translation pairs	Raw natural text
Purpose	Evaluation benchmark	Machine translation	Language modeling ✓
Has Guarani?	Yes (but translated)	No	Yes (original) ✓
Usable as training?	Explicitly says no	Wrong task type	Yes ✓
Fits this project?	No	No	Yes ✓

What changed after investigating FLORES+

After realizing FLORES+ was not suitable for training, I needed a real text corpus. I searched for a public Guarani dataset and found Guarani Wikipedia.

New training dataset: Guarani Wikipedia

Field	Value
Dataset	wikimedia/wikipedia
Subset	20231101.gn
Language	Paraguayan Guarani
Rows before cleaning	5,519
Rows after cleaning	3,535
Useful columns	id, url, title, text

Why I chose it

Guarani Wikipedia contains real Guarani text and is better suited for language model fine-tuning than a benchmark-style dataset. The size (after cleaning) falls within the expected low-resource range of 1,000 to 5,000 examples. It can be loaded with the Hugging Face datasets library and cleaned in Python.

Evidence from my test

I loaded the dataset in Python and inspected sample rows. The dataset loaded successfully and contained real Guarani Wikipedia article text. Cleaning reduced rows from 5,519 to 3,535, filtering out missing, empty, very short, and duplicate texts. The larger-than-expected reduction makes sense because Wikipedia contains many short stubs that do not provide enough language context for fine-tuning.

Dataset size decision

I will use 3,000 examples from the cleaned 3,535. This is large enough to give the model meaningful language patterns, but controlled enough for the available time and hardware.

Final Dataset Strategy

Role	Dataset
Training	Guarani Wikipedia (wikimedia/wikipedia, subset 20231101.gn)
Validation	Held-out split from Guarani Wikipedia
Test	Held-out split from Guarani Wikipedia
Optional evaluation reference	FLORES+ (not training data)

Important limitations

- Paraguayan Guarani, not Guarani Mbya: This is a deliberate and responsible choice. Using sensitive Indigenous community data without consent would be an ethical problem. Guarani Mbya is the inspiration and ethical context for this project, not the technical target. The core technical question, whether a small model fine-tuned on limited data can generalize rather than memorize, does not depend on the specific language. Paraguayan Guarani is a suitable and safe stand-in for testing that question.
- Wikipedia text is encyclopedic, not natural writing assistant conversation text.
- I do not speak the language, so I cannot judge linguistic quality.

My evaluation will therefore focus on technical behavior, memorization, generalization, and output comparison, rather than language correctness.